

$$(i) \begin{bmatrix} 3 & 5 \\ 1 & -1 \end{bmatrix} \quad (ii) \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix} \quad (iv) \begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix}$$

Choose the correct answer in the Exercises 11 and 12.

11. If A, B are symmetric matrices of same order, then $AB - BA$ is a

- (A) Skew symmetric matrix (B) Symmetric matrix
(C) Zero matrix (D) Identity matrix

12. If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, and $A + A' = I$, then the value of α is

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
(C) π (D) $\frac{3\pi}{2}$

3.7 Invertible Matrices

Definition 6 If A is a square matrix of order m , and if there exists another square matrix B of the same order m , such that $AB = BA = I$, then B is called the *inverse* matrix of A and it is denoted by A^{-1} . In that case A is said to be invertible.

For example, let

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \text{ be two matrices.}$$

Now

$$\begin{aligned} AB &= \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -3 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & -6+6 \\ 2-2 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \end{aligned}$$

Also

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \text{ Thus B is the inverse of A, in other}$$

words $B = A^{-1}$ and A is inverse of B, i.e., $A = B^{-1}$

Note

1. A rectangular matrix does not possess inverse matrix, since for products BA and AB to be defined and to be equal, it is necessary that matrices A and B should be square matrices of the same order.
2. If B is the inverse of A , then A is also the inverse of B .

Theorem 3 (Uniqueness of inverse) Inverse of a square matrix, if it exists, is unique.

Proof Let $A = [a_{ij}]$ be a square matrix of order m . If possible, let B and C be two inverses of A . We shall show that $B = C$.

Since B is the inverse of A

$$AB = BA = I \quad \dots (1)$$

Since C is also the inverse of A

$$AC = CA = I \quad \dots (2)$$

Thus

$$B = BI = B(AC) = (BA)C = IC = C$$

Theorem 4 If A and B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Proof From the definition of inverse of a matrix, we have

$$(AB)(AB)^{-1} = I$$

$$\text{or} \quad A^{-1}(AB)(AB)^{-1} = A^{-1}I \quad (\text{Pre multiplying both sides by } A^{-1})$$

$$\text{or} \quad (A^{-1}A)B(AB)^{-1} = A^{-1}I \quad (\text{Since } A^{-1}A = I)$$

$$\text{or} \quad IB(AB)^{-1} = A^{-1}I$$

$$\text{or} \quad B(AB)^{-1} = A^{-1}I$$

$$\text{or} \quad B^{-1}B(AB)^{-1} = B^{-1}A^{-1}I$$

$$\text{or} \quad I(AB)^{-1} = B^{-1}A^{-1}I$$

$$\text{Hence} \quad (AB)^{-1} = B^{-1}A^{-1}$$

EXERCISE 3.4

1. Matrices A and B will be inverse of each other only if

(A) $AB = BA$ (B) $AB = BA = 0$

(C) $AB = 0, BA = I$ (D) $AB = BA = I$